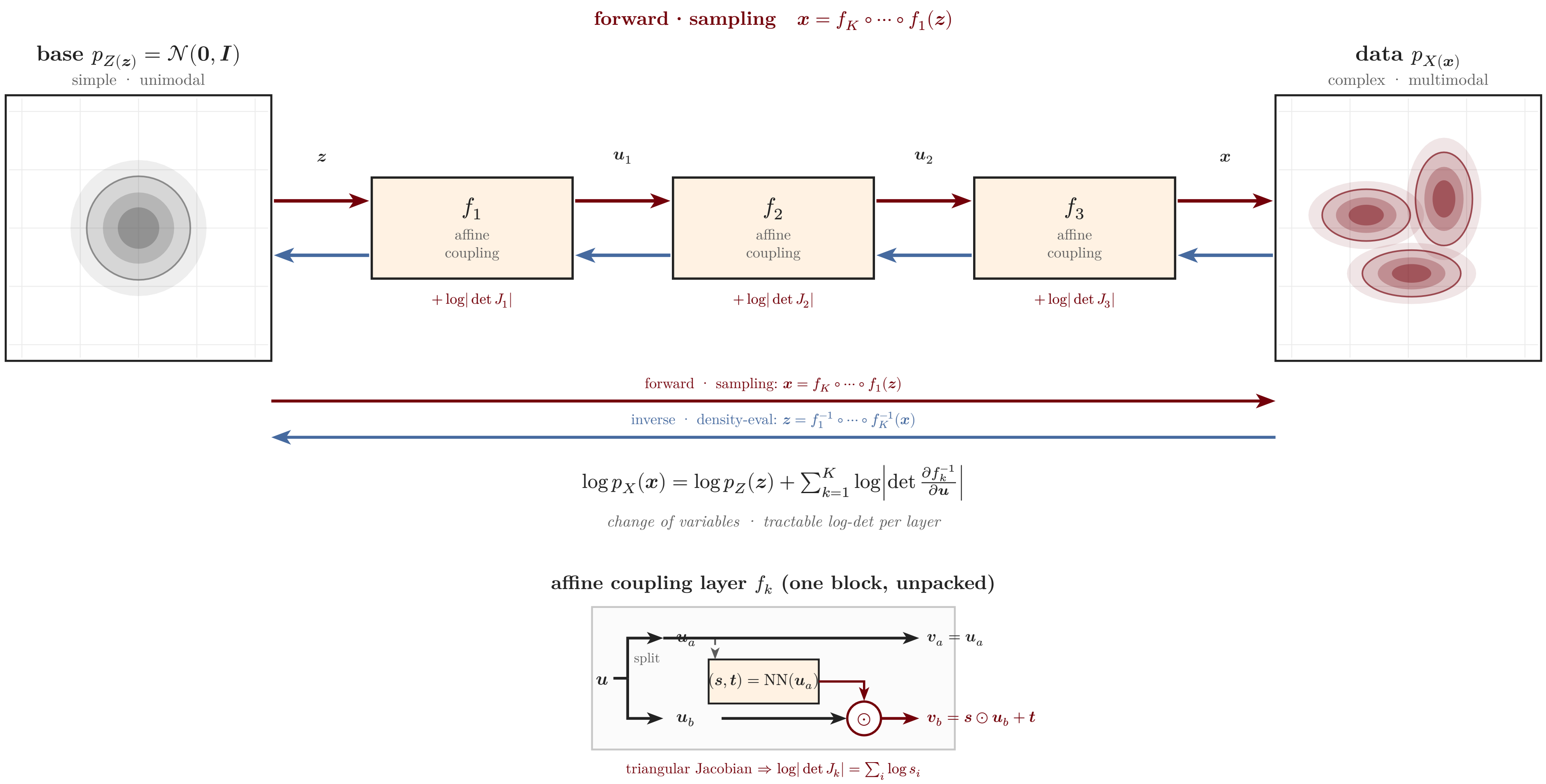

forward · sampling $x = f_K \circ \cdots \circ f_1(z)$
base $p_{Z(z)} = \mathcal{N}(\mathbf{0}, \boldsymbol{I})$
simple · unimodal
data $p_{X(x)}$
complex · multimodal
z
u_1
u_2
x
f_1 affine coupling
f_2 affine coupling
f_3 affine coupling
$+\log|\det J_1|$
$+\log|\det J_2|$
$+\log|\det J_3|$
forward · sampling: $x = f_K \circ \cdots \circ f_1(z)$
inverse · density-eval: $z = f_1^{-1} \circ \cdots \circ f_K^{-1}(x)$
$\log p_X(x) = \log p_Z(z) + \sum_{k=1}^{K} \log\left|\det \frac{\partial f_k^{-1}}{\partial u}\right|$
change of variables · tractable log-det per layer
affine coupling layer f_k (one block, unpacked)
u
split
u_a
u_b
$(s, t) = \mathrm{NN}(u_a)$
$\odot$
$v_a = u_a$
$v_b = s \odot u_b + t$
triangular Jacobian $\Rightarrow \log|\det J_k| = \sum_i \log s_i$